

DATA620013 Advanced Statistical Learning

Assignment 2

Date: 2023-12-04

Exercise 1.

Natural Cubic Splines. Consider the truncated power series representation for cubic splines with K interior knots. Let

$$f(X) = \sum_{j=0}^3 \beta_j X^j + \sum_{k=1}^K \theta_k (X - \xi_k)_+^3. \quad (1)$$

Prove that the natural boundary conditions for natural cubic splines (Section 5.2.1) imply the following linear constraints on the coefficients:

$$\beta_2 = 0, \sum_{k=1}^K \theta_k = 0; \quad \beta_3 = 0, \sum_{k=1}^K \xi_k \theta_k = 0. \quad (2)$$

Hence derive the basis

$$N_1(X) = 1, \quad N_2(X) = X, \quad N_{k+2}(X) = d_k(X) - d_{K-1}(X), \quad (3)$$

where

$$d_k(X) = \frac{(X - \xi_k)_+^3 - (X - \xi_K)_+^3}{\xi_K - \xi_k}. \quad (4)$$

Exercise 2.

Given data y_i with mean $f(x_i)$ and variance σ^2 , and a fitting operation $\mathbf{y} \rightarrow \hat{\mathbf{y}}$, let's define the degrees of freedom of a fit by $\sum_i \text{cov}(y_i, \hat{y}_i)/\sigma^2$. Consider a fit $\hat{\mathbf{y}}$ estimated by a regression tree, fit to a set of predictors $X_1, X_2, \dots, X_p$.

- (1) In terms of the number of terminal nodes m , give a rough formula for the degrees of freedom of the fit.
- (2) For a linear smoother $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$, show that

$$\sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i) = \text{trace}(\mathbf{S})\sigma_\varepsilon^2$$

- (3) For linear regression predictor $\hat{\mathbf{f}} = \mathbf{S}\mathbf{y} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$, if S_{ii} is the i th diagonal element of $\mathbf{S}$, $\hat{f}^{-i}(x_i)$ is the leave- i th-out prediction of x_i , show that

$$y_i - \hat{f}^{-i}(x_i) = \frac{y_i - \hat{f}(x_i)}{1 - S_{ii}}$$

Hint: Using Sherman–Morrison–Woodbury formula

$$(A - bb^T)^{-1} = A^{-1} + A^{-1}b \frac{1}{1 - b^T A^{-1}b} b^T A^{-1}$$

Exercise 4. Programming: EM Algorithm for Two-component Gaussian Mixture.

According to the following Table 1 data (see attachment data02.txt), write a program to implement Algorithm 8.1 in Section 8.5.1 of the textbook, and calculate the maximum likelihood estimation of the corresponding parameters (μ, σ, π) . Select some iterations to list $\hat{\pi}$ (as TABLE 8.2), and draw the observation values and Gaussian mixture density function plots corresponding to different iterations. Finally, plot the log-likelihood against the number of iterations following FIGURE 8.6.

1.37	0.77	1.91	1.68	2.51	1.75	0.18	1.03	2.72	2.62	1.65	1.21	4.88	4.45	4.53
4.67	5.67	4.46	4.40	4.22	6.04	4.67	5.81	5.79	5.09	4.22	5.76	5.41	3.57	5.74

Table 1: 30 fictitious data points used in estimation of the two-component Gaussian mixture.

Notes:

1. It's recommended to use L^AT_EX to complete the assignment, and the document should be submitted in PDF format with the naming format of "Assignment2-张

三-23210980001”. The assignment template can be found in this document. Please indicate your name, student number and date at the beginning of the document.

Assignments are due in two weeks, so please be aware of the deadlines and submit them on eLearning on time!

2. The recommended programming language is Python or R with necessary comments. The attached code should be named in the format of "Assignment2-Code-张三-23210980001", or you may combine the code, documentation and results into one.
3. Please complete the assignment independently, discussion is allowed but plagiarism is forbidden. If you have any questions about the assignments, you may contact the TA.